

DATA ANALYST INTERNSHIP

Task 1: Data Cleaning and Preprocessing

Prepared By: KUNGUMAYEE M

Date: 16.06.2026

1. Project Overview

The objective of this task was to clean and preprocess a raw dataset containing missing values, inconsistent formats, and potential duplicate records. The Netflix Movies and TV Shows dataset was selected for this project due to its diverse categorical and date-related attributes.

2. Dataset Information

Dataset Name: Netflix Movies and TV Shows Dataset

Source: Kaggle

File Format: CSV

3. Data Cleaning Activities Performed

a) Missing Value Treatment

The dataset was examined for missing values using Pandas functions. Missing values were identified in columns such as director, cast, country, rating, duration, and date_added.

Actions Taken:

- Missing values in categorical columns were replaced with “Unknown”.
- Missing values in the date_added column were handled using forward-fill techniques.
- A final validation check was performed to ensure data completeness.

b) Duplicate Record Handling

The dataset was checked for duplicate records using the duplicated() function.

Actions Taken:

- Duplicate rows were identified.

- Duplicate records were removed using `drop_duplicates()` to ensure data integrity.
- c) Data Standardization

Text-based columns were standardized to improve consistency.

Actions Taken:

- Removed leading and trailing spaces.
 - Converted text values into a consistent title-case format.
 - Standardized values across country, rating, and type columns.
- d) Date Format Standardization

The `date_added` column contained date information that required formatting.

Actions Taken:

- Converted values into datetime format.
 - Standardized dates into DD-MM-YYYY format.
- e) Column Name Standardization

Column names were cleaned to improve readability and maintain coding standards.

Actions Taken:

- Converted all column names to lowercase.
 - Replaced spaces with underscores.
- f) Data Type Validation

Data types were reviewed and validated.

Actions Taken:

- Verified data types using `df.dtypes`.
 - Ensured `release_year` was stored as an integer data type.
- g) Outlier Analysis

Outlier detection was performed on the `release_year` column using the Interquartile Range (IQR) method.

Actions Taken:

- Identified records outside the acceptable range.

- Outliers were documented for further analytical consideration.

4. Tools Used

- Python
- Pandas
- Visual Studio Code

5. Deliverables

The following files were generated as part of the project:

- Original Dataset (netflix_titles.csv)
- Cleaned Dataset (cleaned_netflix_titles.csv)
- Python Source Code (task1.py)
- README Documentation
- Project Screenshots

6. Outcome

The dataset was successfully cleaned, standardized, and transformed into an analysis-ready format. The resulting dataset is suitable for further exploratory data analysis, visualization, and machine learning applications.

Conclusion

This project provided practical experience in handling real-world data quality issues, including missing values, duplicate records, inconsistent formats, and data validation. The task strengthened proficiency in Python, Pandas, and data preprocessing techniques commonly used in data analytics projects.